TCD: Task-Collaborated Detector for Oriented Objects in Remote Sensing Images

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***Abstract*— Oriented object detection (OOD) in remote sensing image interpretation is challenging due to the difficulty of locating objects with arbitrary orientations. Existing methods have made considerable progress based on oriented heads or anchors. However, most of them follow the classical detection paradigm, such as assigning samples based on Intersection-over-Unions (IoU) and predicting through two independent tasks. These fixed strategies impair the consistency between classification and localization predictions, resulting in the prediction with optimal localization accuracy being suppressed by the nonoptimal ones during nonmaximum suppression (NMS). To address this problem, a task-collaborated detector (TCD) is proposed. Compared with current single-stage methods, its improvements include two aspects: task-collaborated assignment (TCA) and task-collaborated head (TCH). Specifically, to better pull closer the best anchors for two tasks, TCA introduces classification and localization confidence into sample assignment and tends to select the anchors with accurate and consistent predictions as positive during training. TCH provides a better balance for learning interactive and discriminative features. It can flexibly adjust the spatial feature distribution of classification and localization tasks by learning the joint features from the aggregation layer. Extensive experiments are conducted on HRSC2016, DOTA, and DIOR-R, and the proposed TCD achieves the state-of-the-art performance [90.60, 80.89, and 65.04 mean average precision (mAP), respectively]. Consistency analysis also demonstrates that TCD can significantly improve prediction consistency.**

***Index Terms*— Label assignment, oriented object detection (OOD), remote sensing images, task-collaborated detector (TCD).**

I. Introduction

ORIENTED detection, which can offer objects' orientation and scale information, is widely employed in remote sensing image interpretation. Recently, dense object detectors [1], [2], [3], [4], [5] become increasingly popular due to their elegant and effective framework, which greatly promotes the development of oriented object detection (OOD). There are many approaches that follow the route of anchor-based detector [3], [6], [7] and have achieved satisfactory performance using additional orientation prediction [8], [9] or oriented anchors [10], [11], [12]. Most of these methods follow the fixed paradigm in natural image detection, such as constant Intersection over Union (IoU)-based label assignment, independent classification and localization tasks, etc., which leads to the inconsistency of the predictions of two tasks (classification and localization) to a large extent. As shown in Fig. 1 (top), the high localization confidence [IoU between ground-truth (GT) and predicted oriented boxes] of the predicted boxes corresponds to low classification confidence; therefore, the prediction with high localization confidence may be suppressed during nonmaximum suppression (NMS). The main factors that lead to the inconsistency are as follows.

1) *Fixed Label Assignment:* Anchor-based oriented methods expect that anchors can list all possible locations, scales, or orientations for objects and can be used as regression or classification candidates. They rely on

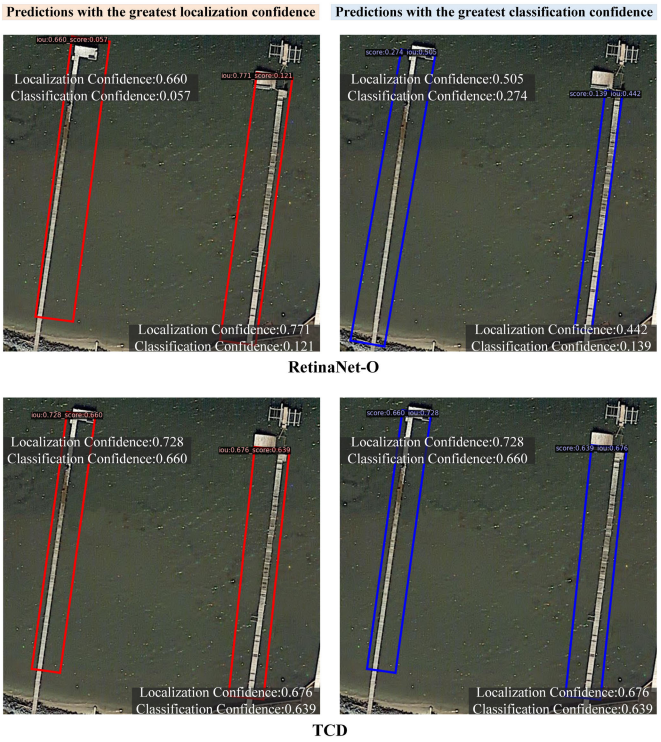


Fig. 1. Visualization of some detection results predicted by (Top) RetinaNet-O [3] and (Bottom) TCD. The predicted boxes in the left column have the largest localization confidence, while those in the right column have the best classification confidence. As we can see, the optimal predictions in RetinaNet-O are suppressed by the nonoptimal ones due to the ranking of the classification confidence during NMS.

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the IoUs between GT and preset anchors to assign samples, which assumes that anchors with high IoUs are appropriate for localization and classification. However, in reality, this fixed paradigm does not necessarily match the optimal anchor for oriented objects. One of the main obstacles is that the constant IoU threshold (0.5 in RetinaNet [3]) for positive/negative is not friendly to long objects. These arbitrary-oriented objects generally have a small IoU with the preset anchors, which prevents them from getting enough samples to participate in the training. IoU-based assignment schemes depend too much on the overlaps between anchors and GT, but the final performance is not entirely dependent on the degree of overlap between anchors and GT [13]. Similar to the calculation of IoU, most anchor free methods [4], [5] use a geometric approach to select anchor points close to the center of the object. However, these strategies are insensitive to the object's context and there are differences in optimal anchors for classification and localization [14]. Whether anchor-based or anchor-free detectors, these widely used assignment methods cannot collaborate with two tasks, thus making it difficult for detectors to make accurate and consistent predictions for the two tasks.

2) *Independence of Classification and Localization Tasks:* Recent detectors usually optimize classification and localization tasks in the form of multitask joint training [3], [4], [5], [8], [12], which assumes that the two branches have the consistent prediction. However, classification and localization are trained independently. They have their losses and solve different problems in such detection pipelines. Although some works [15], [16], [17] try to improve the consistency of predictions in terms of feature alignment, they still maintain independent structures. Such an independent design might cause a lack of interaction between the two tasks, leading to an inconsistency in predictions. Moreover, the classification label is only 0 and 1, indicating whether it belongs to the background or foreground. Due to the annotations for the classification task in object detection are the boxes in the image, there is no suitable metric to indicate the degree of overlap between anchors and objects during training the classifier. For example, if there are five anchors, and their IoU with an object are 0.5, 0.6, 0.7, 0.8, and 0.9, respectively, then they are all assigned as positive. But apparently the anchor with IoU equal to 0.9 is closest to the real object. These factors result in a gap between classification and localization. Therefore, bounding boxes with high localization accuracy may be eliminated by lower ones during NMS.

To illustrate the above problems more intuitively, we visualize the correlation between classification and localization confidence by selecting top eight predictions (top eight localization confidence) for each instance before NMS on DOTA [18] dataset. As shown in Fig. 2, the horizontal axis represents the localization confidence, while the vertical axis denotes its classification confidence. As can be seen from Fig. 2(a), most predictions with high localization confidence tend to have a low classification score in RetinaNet-O [3]. The Pearson correlation coefficient (PCC) also shows that the localization confidence is not consistent with the classification confidence in RetinaNet-O.

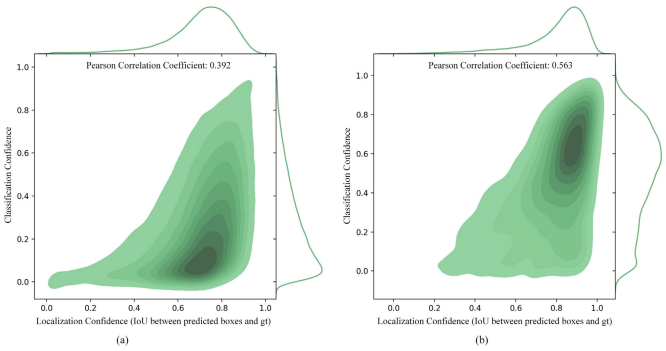


Fig. 2. Correlation between classification and localization confidence of top eight predictions. (a) RetinaNet-O. (b) TCD.

In this article, a task-collaborated detector (TCD) is proposed to alleviate the inconsistent problem, expecting to obtain more consistent predictions. First, a task-collaborated assignment (TCA) scheme is designed to collaborate with two tasks instead of relying on IoU alone. The anchors tend to obtain accurate and consistent predictions will be more inclined to be selected as positive. This collaborated scheme can not only dynamically select high-quality samples for two tasks but also explicitly close the distance between the input anchors and the output localization/classification confidence. Next, a task-collaborated head (TCH) is designed to replace the original two independent branches. Compared with the conventional parallel head, TCH enhances the interaction of the two tasks through the feature aggregation layer and dynamically adjusts the feature spatial distribution through the consistency weight layer. With the help of task-collaborated loss (TCL), the two tasks in TCH easily adjust the feature distribution, allowing the two tasks to find more consistent features and further collaborate the two tasks on prediction. TCL makes the classifier more sensitive to potential anchors with high localization confidence and focuses more on inconsistent predictions. Thus, the consistent prediction (a box with high localization confidence and jointly having the high classification confidence) can be preserved during inference.

The main contributions of this article are as follows.

1) This article analyzes the reasons for inconsistent predictions from two aspects (the label assignment scheme and the independence of head) and then proposes a task-collaborated oriented detector to make predictions with high quality in both classification and localization.
2) The TCA explicitly collaborates the two tasks in selecting more task-collaborated anchors, and the high-quality (i.e., tend to get consistent and accurate predictions) anchors are easier to be assigned as positive samples.
3) The TCH is more interactive than independent branches, and with the help of TCL, it is easier to adjust the feature distribution so that the two tasks can collaborate on prediction.

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II. Related Work

A. Oriented Object Detection

In the past few years, remarkable progress [2], [3], [6], [19], [20], [21], [22] have been made in natural object detection, which brings new vitality to object detection in remote sensing images [9], [23], [24], [25], [26]. Arbitrary OOD, as a research hotspot, has been extensively studied based on the efficient detection frameworks in natural images. Most of them [8], [9], [10], [11], [25], [26], [27], [28] achieve orientation prediction by adding angular regression. Setting anchors with various angles is a conventional solution for oriented detectors, which can better cover oriented objects. Rotation region proposal networks (RRPNs) [10] and refined rotation retinanet detector (R3Det) [12] apply oriented anchors directly to the image, then produce oriented region proposals or predictions. Although they achieve acceptable performance, the computational cost is also greatly increased, for which RoI-Transformer [26] that can automatically generate rotated candidates from horizontal proposals is proposed to reduce the number of oriented anchors. Recently, Oriented regions with CNN features (R-CNN) [29], anchor-free oriented proposal generator (AOPG) [30], and anchor-free oriented region proposal network (AFO-RPN) [31] directly generate high-quality rotated candidates without enormous predefined anchors, which can greatly improve detection performance while maintaining speed. Moreover, some methods do not directly regress the orientation of the objects. Gliding vertex [32] demonstrates a new oriented bounding box representation by predicting four-point polygons based on the horizontal bounding box. Oriented objects detection network (O2-DNet) [33] and polar remote sensing object detector (P-RSDet) [34] are based on key points. The former uses the center point and two center lines to represent the rotated box, and the latter uses the center point, angle in polar coordinates, and radius to construct the oriented bounding box. Circular smooth label (CSL) [35] uses a CSL to solve the loss mutation caused by the periodic change in angle regression. Densely coded labels (DCL) [36] is optimized for the problems that the head is too heavy and the detection of square-like objects is not friendly to CSL. Its accuracy and speed are significantly improved.

B. Label Assignment

Conventional object detectors [2], [3], [6] select samples by computing the IoU between anchors and GT, while Cascade R-CNN [19] pointed out that training with a single IoU threshold lead to degraded performance on other IoUs. Therefore, Cascade R-CNN [19] selects high-quality samples by gradually increasing the threshold in several stages but brings about a reduction in the efficiency. Recently, anchor-free detectors fully convolutional one-stage (FCOS) [4] and Foveabox [37] directly assign positive/negative samples based on the spatial distance. Although anchor-based and anchor-free detectors differ in many aspects, they adopt a fixed strategy (IoU threshold or fixed region) for sample assignment. Therefore, many works in recent years have attempted to make sample assignment more adaptable to improve the network's learning ability for high-quality samples further. Adaptive training sample selection (ATSS) [5] uses an adaptive assignment method that takes the sum of the mean and standard deviation of the IoU of a set of nearest anchors as the positive/negative threshold for each GT. Dynamic R-CNN [38] automatically adjusts the IoU threshold based on the proposals distribution in the training procedure to select high-quality positive samples. To solve the noisy information brought by binarized sample assignment, Noisy Anchors [39] adopts soft label and anchor reweighting mechanism based on classification and localization loss. Guided Anchoring [40] and Meta Anchor [41] use dynamic anchors to better adapt to various distributions. Free Anchor [42] builds K candidate anchors for each GT, then performs positive and negative sample assignment in each candidate set by maximum likelihood estimation. Probabilistic anchor assignment (PAA) [43] adaptively assigns anchors to positive or negative by fitting the scores of the anchors with probability distributions. Sparse label assignment (SLA) [44] selects high-quality sparse anchors based on the posterior IoU of detections. Dynamic anchor learning (DAL) [45] introduces matching degree in label assignment to measure the localization potential of anchors, which is beneficial to higher quality detection.

C. Feature Alignment Between Classification and Localization

Recently, general object detectors usually consist of two parallel heads to disentangle two tasks: classification and localization. And the classification score plays a key role in postprocessing (e.g., NMS), which, as discussed above, may suppress predictions with high localization confidence. To solve this problem, IoU-Net [16], IoU-aware [15], and PAA [43] adopt an extra branch to predict the localization confidence, and then combine localization confidence with classification confidence as the final detection score. Generalized focal loss (GFL) [17] and VarifocalNet (VFNet) [46] reformulate a joint representation that incorporates classification and localization confidence to eliminate inconsistencies between training and inference. Task-aware spatial disentanglement (TSD) [47] adopts the module of TSD, which allows classification and localization to learn the adaptive proposals and features, respectively. Double Head R-CNN [48] and you only look once X (YOLOX) [49] also use different head structures for different tasks. Generalized focal loss V2 (GFLv2) [50] introduces statistics of bounding box distributions to predict localization scores. Task-aligned one-stage object detection (TOOD) [51] demonstrates an effective head structure to enhance the interaction of classification and localization tasks.

III. Proposed Method

This section will introduce the specific structure of TCD. Similar to most one-stage detectors, ResNet [52] with feature pyramid network (FPN) is adopted to extract multiscale features. We first describe the TCA in Section III-A. The structure of TCH is given in Section III-B, and the TCL is presented in Section III-C. The overall network structure of TCD is shown in Fig. 3.

A. Task-Collaborated Assignment

Due to the overhead view, there are few scenes where objects overlap in remote sensing images. If an anchor

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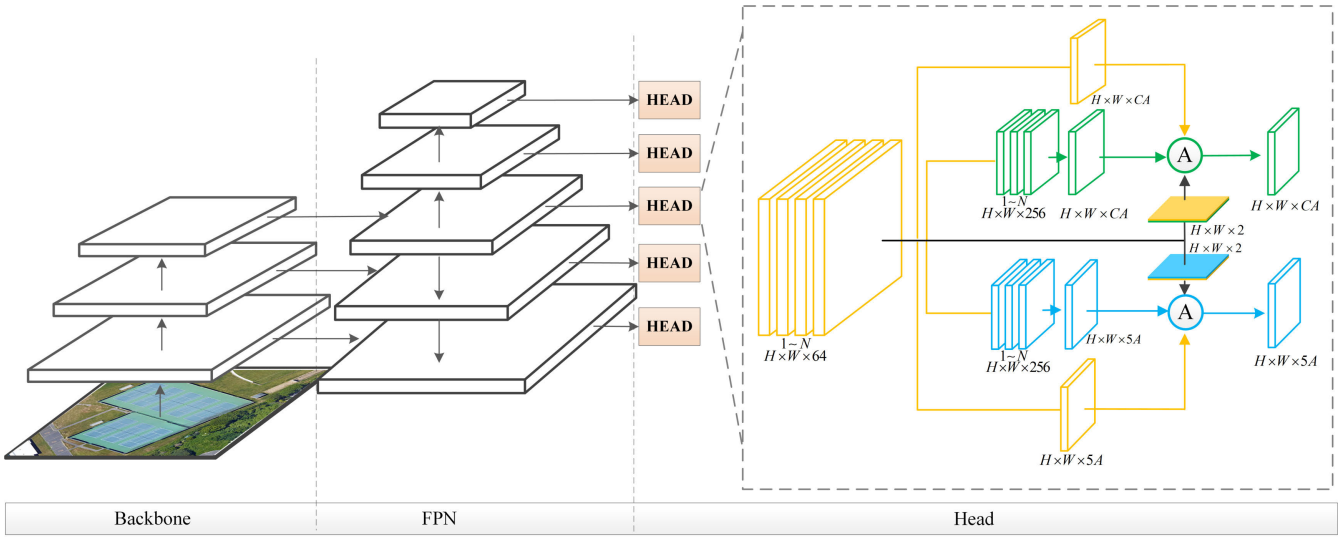


Fig. 3. Structure of the proposed TCD. Similar to RetinaNet [3], TCD is composed of the backbone with FPN and TCH. TCH can make the two tasks more interactive, and its outputs serve as the basis for label assignment to select more consistent anchors. Ⓐ represents the multiply-then-add operator in (10).

can accurately locate and classify (high classification and localization confidence) an object, it must contain a critical semantic area of this object. Therefore, the IoUs between anchors and GT (named input IoU in DAL [45]) are an essential basis for assigning positive and negative samples. However, existing researches show that the distinguishing features needed for localization are often unevenly distributed on GT box [42], [47], and the features required for localization and classification are different [47], [48]. Thus, the assignment based on input IoU only will lead to the inability to obtain critical features for localization and classification. Therefore, we introduce both localization and classification tasks into the sample assignment and design a new assignment metric, i.e., the degree of the two tasks' collaboration of each anchor. Due to the existence of NMS, anchors with greater collaborative should be able to predict high classification and localization confidence simultaneously. Low-quality anchors are just the opposite. Based on the above analysis, a task-collaborated factor (tcf) using the combination of classification and localization confidence is designed to measure the degree of collaborative. It is defined as follows:

$$\text{tcf} = \frac{\beta - (c \times l)^{\alpha}}{\beta} \tag{1}$$

where c and l represent the classification and localization confidence, respectively. β and α are hyperparameters designed to adjust the impact of two tasks in the assignment. Then the tcf and IoU between anchors and object are combined as the metric for anchor assignment, as follows:

$$m = \text{IoU}^{\text{tcf}} \tag{2}$$

$$\text{IoU} = \frac{\text{Area}(A \cap G)}{\text{Area}(A \cup G)} \tag{3}$$

where IoU denotes the IoU (the area of the intersection of the two boxes divided by the area of the union of the two boxes) between anchors and GT. It is worth noting that IoU, tcf $\in [0, 1]$ and the smaller (larger) tcf corresponds to the larger (smaller) m. The smaller tcf means that $c \times l$ needs to be larger. As shown in (1), when c and l are consistent and both large, $c \times l$ is larger. Therefore, there is a strong correlation between m and the consistency of the two tasks' outputs. tcf can continuously optimize m through the output of two tasks, dynamically selecting high-quality (i.e., tend to get consistent and accurate predictions) anchors. We adopt a simple yet effective rule top k to assign positive/negative samples with the newly defined assignment metric. For each GT, k anchors with the highest m values are assigned as positive, and the other anchors as negative. Algorithm 1 shows how the TCA assigns samples.

Algorithm 1 TCA

Input:

G_b: the list of all GT boxes on input image.
G_c: the list of all GT categories corresponding to G_b.
B: the list of all predicted oriented bounding boxes.
S: the list of all predicted classification scores.
A: the list of all anchors.
k, α, β are hyperparameters and default to $8, 0.5, 2$

Output:

P: the list of positive anchors
N: the list of negative anchors

for each GT box, GT category $g \in G_b$, $c \in G_c$ **do**
  calculate IoU between g and A: $IoU_{anchor} = IoU(g, A)$;
  calculate IoU between g and B: $IoU_{pred} = IoU(g, B)$;
  compute classification scores for category c: $S_c = S[c]$;
  compute task-collaborated factor tcf and assignment metric m: $tcf = \frac{\beta - (S_c \times IoU_{pred})^{\alpha}}{\beta}$, $m = IoU_{anchor}^{tcf}$
  select top k anchors as positive samples based on m: $ind \leftarrow$ indexes of the largest k values in m $P = P \cup A[ind]$
end for
assign negative samples: $N = A - P$
return P, N

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B. Task-Collaborated Head

We expect to design a more interactive head to enhance the connection between two tasks to obtain consistent localization and classification confidence. To achieve this goal, TCH, as shown in Fig. 3 ("Head"), is proposed. It is a simple but high-performance structure that can achieve significant performance improvement while keeping the same inference time as RetinaNet [3]. The structure is described in detail below.

When the FPN features (denote as $F^{\text{fpn}} \in \mathbb{R}^{H\times W\times C}$) are extracted, a feature aggregation layer is introduced to learn the joint features of the two tasks instead of directly learning the separation features of two tasks like RetinaNet [3]. As shown by the orange part in Fig. 3 ("Head"), it adopts N convolutional layers and nonlinear activation functions to enhance two tasks' interaction, and the aggregation of multiple convolutional layers can provide multiple level's features for following classification and localization. It can be formulated as follows:

$$F_i^{\text{aggr}} = \begin{cases} \sigma_i\left(\text{conv}_i\left(F^{\text{fpn}}\right)\right) & i = 1 \\ \sigma_i\left(\text{conv}_i\left(F_{i-1}^{\text{aggr}}\right)\right) & i > 1 \end{cases} \quad \forall i \in \{1, 2, \ldots, N\} \tag{4}$$

$$F^{\text{aggr}} = \text{Cat}\left\{F_1^{\text{aggr}}, F_2^{\text{aggr}}, \ldots, F_N^{\text{aggr}}\right\} \tag{5}$$

where conv_i and σ_i represent the ith conv layer and the relu function, respectively. $\text{Cat}(\cdot)$ means the feature concatenation. Then the aggregated features are fed into the classification and localization branch of two tasks.

Due to the conflict of features required for classification and localization [48], a single branch cannot meet the feature differentiation requirements of two tasks. Consequently, two different branches extract distinguishable features of two tasks. As shown in Fig. 3, the green and blue parts in "Head" represent the classification and localization branches, respectively. The two identical branches are composed of N convolutional layers and one prediction layer, which also maintains the design of RetinaNet [3]

$$F^{\text{score/bbox}} = \sigma_N(\text{conv}_N(\ldots\sigma_1(\text{conv}_1(F^{\text{aggr}}))\ldots)) \tag{6}$$

where conv_i and σ_i represent the ith conv layer and the relu function, respectively. $F^{\text{score/bbox}}$ represents the features for classification and localization. Then the results of classification and localization can be predicted from $F^{\text{score/bbox}}$

$$\begin{aligned} F_C &= \text{conv}(F^{\text{score}}) \\ F_B &= \text{conv}(F^{\text{bbox}}) \end{aligned} \tag{7}$$

where $F_C \in \mathbb{R}^{H\times W\times CA}$ and $F_B \in \mathbb{R}^{H\times W\times 5A}$ represent classification scores and oriented bounding boxes. C denotes the number of classes and A represents the number of anchors on each feature point.

Since two parallel branches bring about independent features, the joint features from the aggregation layer F^{aggr} are adopted to adjust the spatial distributions and enhance the interactive of the two tasks. As shown in Fig. 3, the upper and lower two orange branches correspond to the two tasks of classification and localization, respectively. Specifically, for classification task, classification maps $M_C \in \mathbb{R}^{H\times W\times CA}$ are used to adjust the classification prediction F_C

$$M_C = \text{conv}(F^{\text{aggr}}) \tag{8}$$

where conv denotes the ordinary convolutional layer, and the localization task is the same, localization maps $M_B \in \mathbb{R}^{H\times W\times 5A}$ are also learned from aggregate features and used to adjust the predicted bounding box F_B at each feature point

$$M_B = \text{conv}(F^{\text{aggr}}). \tag{9}$$

Then two spatial consistency weight layers ($W_C, W_B \in \mathbb{R}^{H\times W\times 2}$) are computed from the aggregate features and adjust the spatial feature distribution for each task through the feature aggregation layer. W have two layers (i.e., $W_C[1]$ and $W_C[2]$), corresponding to F and M, respectively. Ⓐ represents the multiply-then-add operator. The predictions (P_C for classification and P_B for localization) are as follows:

$$\begin{aligned} P_C &= W_C[1] \times F_C + W_C[2] \times M_C \\ P_B &= W_B[1] \times F_B + W_B[2] \times M_B. \end{aligned} \tag{10}$$

Finally, a TCL that can further guide the training process of TCH is introduced to make consistent predictions.

C. Task-Collaborated Loss

Most detectors use classification and localization confidence obtained from independent training during inference, which leads to inconsistent outputs (e.g., a high classification score with a low box IoU). To alleviate this problem, inspired by GFL [17], we optimize the consistency of predictions by introducing the localization score (i.e., oriented box IoU) into classification loss. Specifically, we use localization confidence l instead of binary labels during training, a softened one-hot form that floats between 0 and 1 as the change of localization confidence. Note that l is the IoU between the predicted oriented bounding box and its corresponding GT and $l \in [0, 1]$. The setting of Focal loss [3] is adopted to mitigate the imbalance problem in the one-stage dense detector during training

$$L_{\text{cls}} = \lambda \sum_{i=1}^{N} \left|l_i^* - c_i\right|^{\gamma} BCE\left(c_i, l_i^*\right) \tag{11}$$

where N denotes the number of samples, i represents the ith sample of all the samples, c_i represents the classification confidence of the ith sample. λ and γ are modulating parameters. l^* is the normalized form of l, as shown in (12). It is used because the value of oriented box IoU score l may be small, which affects the network training

$$l^* = \sqrt{\frac{l}{\max(l)}}. \tag{12}$$

The normalized l has the following three benefits: 1) l^* of the negative sample is set to 0, thereby reducing the prediction scores of the low-quality anchors [e.g., having a small m in (2)]; 2) it guarantees that some difficult samples (positive anchors have small l or m) can effectively participate in training; and 3) it ensures that the localization confidence

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ranking is not changed to optimize the network toward consistent predictions.

As shown in (11), L_{cls} has the minimum value when $l_i^* = c_i$. Similar to Focal loss [3], the focusing factor $|l^* - c|^\gamma$ is relatively large when the classification confidence c deviated away from the label l^* and the loss will pay more attention to optimize this sample. Conversely, when the predictions of two tasks are accurate and consistent, the loss will greatly reduce the weight.

IV. Experiments

A. Dataset and Implementation Details

1) Dataset: HRSC2016[1] [53] is a challenging dataset for ship detection containing objects with different aspect ratios at various angles. The dataset contains 1061 high-resolution images taken from Google Earth and sizes from 300 × 300 to 1500 × 900 pixels. All experimental results are tested on the test set (444 of 1061), and the rest are used to train the TCD.

DOTA[2] [18] is a large-scale dataset for OOD. It has a total of 2806 images collected from various sensors and platforms. There are 188 282 instances divided into 15 categories: helicopter (HC), small vehicle (SV), bridge (BR), plane (PL), ship (SH), harbor (HA), large vehicle (LV), soccer-ball field (SBF), baseball diamond (BD), swimming pool (SP), tennis court (TC), ground track field (GTF), roundabout (RA), basketball court (BC), and storage tank (ST). If not specified, the training and validation sets are used for training, and the official platform is used for testing.

DIOR-R[3] [30] is a new dataset that reannotated oriented bounding box on the original DIOR [24] dataset to adapt arbitrary OOD task. The number of images and categories is consistent with the original data. There are 192 518 instances divided into 20 categories, including bridge (BR), ship (SH), airplane (APL), stadium (STA), tennis court (TC), chimney (CH), overpass (OP), vehicle (VE), baseball field (BF), airport (APO), basketball court (BC), ground track field (GTF), expressway service area (ESA), train station (TS), golf field (GF), dam (DAM), expressway toll station (ETS), storage tank (STO), harbor (HA), and windmill (WM). Unlike DOTA to submit results on the official website, it has released all the annotations. The training and validation sets are used to train, and the testing set is used to evaluate the TCD.

All experiments are based on the standard evaluation protocol of mean average precision (mAP) (VOC2007) [54]. The AP summaries the shape of the precision/recall curve, and is defined as the mean precision at a set of 11 equally spaced recall levels [1, 0, 0.1, ...]

$$\text{AP} = \frac{1}{11} \sum_{r \in \{0, 0.1, \ldots, 1\}} \text{pinterp}(r). \tag{13}$$

The precision at each recall level r is interpolated by taking the maximum precision measured for a method for which the corresponding recall exceeds r

$$\text{pinterp}(r) = \max_{\widetilde{r}: \widetilde{r} \geq r} p\left(\widetilde{r}\right) \tag{14}$$

where $p\left(\widetilde{r}\right)$ is the measured precision at recall $\widetilde{r}$.

2) Implementation Details: The oriented rectangle is defined in the form of (x, y, w, h, θ), where (x, y) represents the center coordinates, w and h represent the length of the long side and short side. θ is the angle between the long side and the positive of the horizontal axis. θ is 0° when the long side is parallel to the horizontal axis and its range of $[-90, 90)$, counterclockwise is negative.

The TCD is implemented based on detectron2.[4] We adopt the ResNet-50-FPN model as the backbone if not separately noted, where the batch size is set to 4 (for HRSC2016 and DIOR-R) or 2 (for DOTA), respectively. Stochastic gradient descent (SGD) optimizes TCD, and the learning rate is 0.0025 (with a batch size of 4) or 0.001 (with a batch size of 2). The momentum and the weight decay are 0.9 and 0.0001, respectively. All models are trained for 12 epochs. It should be noted that the aspect ratio of the anchor is set to {0.2, 0.5, 1}. The hyperparameters α, β in (1) and the λ, γ in (11) are set to 0.5, 2 and 1, 2, respectively. Other parameters are set to the same settings as RetinaNet [3]. For the HRSC2016 dataset, the long side of the image is scaled to 800 while the short side is scaled equally. Rotation augmentation (90°, 180°, and 270°) is applied to all the training data. For the DOTA dataset, the images are scaled to three scales (0.5, 1.0, 2.0) and then split into slices (1024 × 1024 pixels) with 500 pixels overlap. Rotation augmentation is also used for all the training data. We do not scale the image for the DIOR-R dataset but augment the data by flipping and rotating to avoid over-fitting.

B. Comparison With the State-of-the-Art Methods

1) Results on HRSC2016: The dataset contains many ships with large aspect ratios at various angles, which is extremely challenging for performance. The TCD achieves 90.60 mAP while maintaining single-stage detection speed. As shown in Table I, the upper and lower of the table represent two-stage methods and the one-stage methods, respectively. TCD achieves better performance than the two-stage algorithms, which is more obvious than one-stage methods. Fig. 4 shows the visualization of some detection results.

2) Results on DOTA: TCD is compared with several advanced methods on the DOTA test set, the quantitative results as shown in Table II. In general, two-stage methods have more advantages in performance because of their complex model structures. For example, RoI Transformer, AOPG, and Oriented R-CNN perform pretty well. However, when using ResNet-101-FPN as the backbone and multiscale training and testing, TCD achieves the highest performance, even better than the best performance of most two-stage algorithms (1.07, 0.23, and 0.02 mAP higher than RoI Transformer, AOPG, and Oriented R-CNN, respectively). TCD also achieves the best performance in the Basketball court and Bridge categories. Fig. 5 shows the visualization of some detection results on DOTA test set.

3) Results on DIOR-R: Since the DIOR-R dataset has not been released for a long time, there are not many experiments on this dataset. Here are several OOD methods tested in

[1] https://sites.google.com/site/hrsc2016/annotations
[2] https://captain-whu.github.io/DOTA/dataset.html
[3] https://gcheng-nwpu.github.io
[4] https://github.com/facebookresearch/detectron2

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TABLE I
DETECTION RESULTS ON HRSC2016 TEST SET. ∗ REPRESENT MULTISCALE TRAINING

Method	R2CNN [8]	RRPN [10]	RoI-Trans [23]	Gliding [29]	CSL [32]	AOPG [27]	Oriented [26]	Oriented [26]	TCD
Backbone	R-101	R-101	R-101	R-101	R-152	R-50	R-50	R-101	R-101
mAP	73.07	79.08	86.20	88.20	89.63	90.34	90.40	90.50	90.52
Method	Retina-H [12]	RSDet [55]	ARG* [56]	Retina-R [12]	R3Det [12]	RIDet-O [57]	CFC-Net* [13]	S^2Net [58]	**TCD**
Backbone	R-101	R-152	R-101	R-101	R-101	R-101	R-101	R-50	R-50
mAP	82.89	86.5	88.08	89.18	89.26	89.63	89.7	90.17	**90.60**

TABLE II
COMPARISON WITH ADVANCED METHODS ON THE DOTA TEST SET. * DENOTES MULTISCALE TRAINING AND TESTING

Method	Backbone	HC	SV	BR	PL	SH	HA	LV	SBF	BD	SP	TC	GTF	RA	BC	ST	mAP
Two-stage																	
R2CNN [8]	R-101	48.22	59.92	35.34	80.94	55.81	55.14	50.91	55.06	65.67	53.35	90.67	67.44	52.23	66.92	72.39	60.67
RRPN [10]	R-101	53.58	51.85	31.66	88.52	57.25	53.08	56.19	56.69	71.20	51.94	90.81	59.30	52.84	72.84	67.38	61.01
ICN [59]	R-101-FPN	50.23	64.89	47.70	81.36	69.98	67.02	67.82	53.64	74.30	64.17	90.76	70.32	62.90	79.06	78.20	68.16
CAD-Net [22]	R-101-FPN	62.20	71.10	49.40	87.80	76.60	62.00	63.50	48.40	82.40	67.00	90.90	73.50	60.90	79.20	73.30	69.90
SCRDet [9]	R-101-FPN	65.21	68.36	52.09	89.98	72.41	66.25	60.32	65.02	80.65	68.24	90.85	68.36	66.68	87.94	86.86	72.61
Gliding [29]	R-101-FPN	57.32	73.01	52.26	89.64	86.82	72.94	73.14	59.55	85.00	70.86	90.74	77.34	70.91	79.02	86.81	75.02
Mask OBB [60]	X-101-FPN	63.32	76.52	54.21	89.56	85.63	73.94	74.16	54.89	85.95	69.06	89.85	72.90	69.64	83.81	86.48	75.33
CenterMap [61]	R-101-FPN	66.06	77.66	54.60	89.83	87.19	74.13	78.32	56.46	84.41	71.56	90.66	70.25	69.23	84.89	85.27	76.03
CSL* [32]	R-152-FPN	68.93	70.44	54.64	90.25	77.62	73.83	73.51	69.60	85.53	71.10	90.84	75.31	68.04	86.15	86.69	76.17
RoI Trans* [23]	R-50-FPN	73.43	75.22	57.83	87.89	88.04	79.00	84.37	71.04	85.01	**83.29**	90.88	78.55	69.67	87.28	85.79	79.82
DODet [62]	R-50-FPN	**74.47**	78.71	58.01	89.96	88.59	82.06	**85.46**	70.50	85.52	77.43	90.89	81.22	71.54	87.12	87.80	80.62
AOPG* [27]	R-50-FPN	73.10	78.70	60.90	89.88	88.85	82.31	85.29	71.66	85.57	77.32	90.89	81.51	68.69	87.60	87.65	80.66
Oriented* [26]	R-50-FPN	74.11	79.71	61.09	89.84	88.82	**82.42**	85.35	**72.21**	85.43	78.18	90.88	79.82	70.80	86.68	87.73	80.87
One-stage																	
PIoU [63]	DLA-34	64.00	38.30	24.10	80.90	64.80	57.10	64.40	46.50	69.70	61.90	90.90	60.20	37.10	77.20	70.40	60.50
RetinaNet-O [3]	R-50-FPN	40.83	71.06	42.80	86.54	73.53	60.79	58.50	52.42	77.45	64.84	90.72	64.87	62.16	80.97	66.67	66.28
DRN* [24]	H-104	58.48	76.22	47.22	89.71	85.84	69.30	74.43	57.65	82.34	69.63	90.57	64.10	61.93	86.18	84.89	73.23
RSDet [55]	R-152	63.70	70.20	53.80	**90.10**	73.60	66.10	78.70	64.30	82.00	69.30	**91.20**	68.50	68.20	87.10	84.70	74.10
R3Det* [12]	R-152	72.62	78.76	48.11	89.80	87.84	67.53	83.27	65.67	83.77	78.56	90.82	66.77	62.68	85.38	85.51	76.47
DAL [42]	R-50-FPN	64.35	78.30	55.03	89.69	88.46	76.86	81.90	64.41	83.11	72.09	90.89	71.00	65.65	84.97	87.46	76.95
S2A-Net* [58]	R-50-FPN	69.58	79.94	57.74	88.89	**89.11**	78.30	83.19	70.30	83.60	77.01	90.78	**81.95**	68.25	84.87	**87.81**	79.42
TCD	R-50-FPN	70.89	65.75	56.91	89.27	70.67	76.54	76.76	60.16	83.79	71.95	90.88	72.13	71.21	86.95	83.81	75.18
TCD	R-101-FPN	62.27	69.83	54.60	89.60	76.06	77.67	76.61	63.63	85.51	72.00	90.82	75.21	68.87	87.68	85.85	75.75
TCD*	R-50-FPN	71.77	**80.56**	58.18	89.78	88.31	77.84	83.88	68.04	**86.29**	78.92	90.84	78.97	**72.50**	88.01	86.85	80.05
TCD*	R-101-FPN	72.51	80.01	**61.39**	90.04	88.69	80.58	85.28	70.61	86.26	78.97	90.88	79.81	72.44	**88.11**	87.80	**80.89**

TABLE III
COMPARISON WITH ADVANCED METHODS ON THE DIOR-R DATASET. * INDICATES THAT THE RESULTS ARE FROM DODET [62]

Methods	Backbone	BR	SH	APL	STA	TC	CH	OP	VE	BF	APO	BC	GTF	ESA	TS	GF	DAM	ETS	STO	HA	WM	mAP
Faster-O* [6]	R-50-FPN	34.20	81.14	62.79	64.34	81.44	72.57	48.79	50.46	71.72	26.80	80.91	79.24	65.75	47.31	66.63	18.95	66.45	71.21	34.95	65.21	59.54
RetinaNet-O* [3]	R-50-FPN	23.98	77.71	61.49	67.57	81.46	72.54	44.87	38.01	73.57	28.52	81.17	79.54	58.20	47.33	69.25	19.94	72.39	61.09	32.14	60.24	57.55
Gliding* [29]	R-50-FPN	33.88	80.22	65.35	69.26	81.49	74.31	49.64	47.71	74.96	28.87	81.33	78.68	64.70	44.76	72.30	19.58	70.72	61.13	37.22	65.04	60.06
RoI Trans* [23]	R-50-FPN	40.68	81.21	63.34	78.23	81.61	72.60	55.61	43.27	71,78	37.88	87.53	82.74	68.09	54.86	68.96	26.86	78.71	70.26	47.71	65.52	63.87
AOPG* [27]	R-50-FPN	40.90	81.17	62.39	72.69	81.49	72.47	54.45	52.38	71.62	37.79	87.63	81.94	77.99	60.04	73.20	31.08	65.42	71.31	42.32	69.99	64.41
DODet* [62]	R-50-FPN	43.12	85.41	63.40	74.04	81.52	72.59	59.31	51.86	72.11	43.35	81.32	75.47	70.84	55.47	74.15	33.32	78.77	71.56	48.00	66.40	65.10
TCD	R-50-FPN	40.38	85.89	67.95	65.07	81.61	74.74	56.00	48.72	73.48	45.30	83.55	78.22	79.33	57.22	78.03	33.60	70.54	68.99	48.46	63.79	65.04

DODet [62], as shown in Table III. The trainval set is used to train, and the test set is used to evaluate. The TCD achieves 65.04 mAP with ResNet50, which has similar performance to DODet and outperforms other methods (5.5, 4.98, 1.17,

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Fig. 4. Some detection results of TCD on the HRSC dataset.

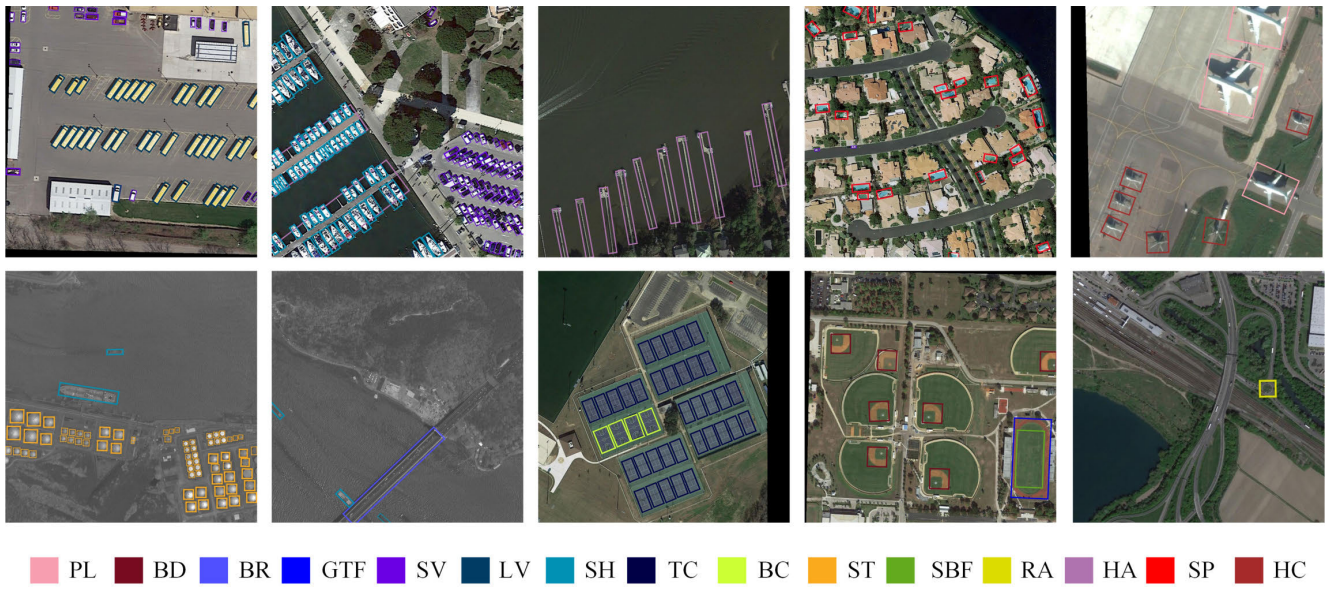


Fig. 5. Some detection results of TCD on the DOTA dataset.

and 0.63 mAP higher than Faster RCNN-O, Gliding Vertex, RoI Trans, and AOPG, respectively). Compared with the one-stage RetinaNet-O, our method achieves a significant performance improvement of 7.49 mAP. The considerable performance advantage verifies the robustness and effectiveness of the TCD. Fig. 6 shows the visualization of some detection results on the DIOR-R test set.

C. Ablation Experiments

In this section, all experiments use the RetinaNet-O [3] with ResNet-50-FPN as the baseline and train the TCD for 12 epochs on the DOTA training set. All experimental results come from the DOTA validation set. To compare with previous methods, only one anchor is set per feature point if not separately stated. AP50 and AP75 represent the mAP at IoU $= 0.5$ and IoU $= 0.75$.

1) Componentwise Analysis: to verify the performance of each component of TCD, componentwise experiments have been conducted on DOTA, as shown in Table IV. First, TCH is compared with the parallel head used in RetinaNet. It is an independent structure and can consistently outperform the parallel head by 1.87–2.88 AP50 (4.20–6.63 AP75) under different assignment schemes with fewer parameters. These results validate that TCH can effectively improve performance by enhancing task interaction and predictive collaboration. Then, the TCA is compared with different assignment schemes, and it improves the performance of over the baseline 3.64 AP50

TABLE IV
PERFORMANCE ON DIFFERENT HEADS AND ASSIGNMENT SCHEMES

Label Assignment	Head	AP50	AP75
RetinaNet-O IoU-based	Original	62.89	28.17
	TCH	65.77	34.80
RetinaNet-O w/ ATSS	Original	64.56	32.21
	TCH	66.87	37.52
RetinaNet-O w/ DAL	Original	65.53	34.55
	TCH	67.40	38.75
RetinaNet-O w/ PAA	Original	65.74	34.02
	TCH	67.81	39.13
RetinaNet-O w/ TCA	Original	66.59	35.79
	TCH	68.75	40.22
	TCH + TCL	**69.86**	**41.30**

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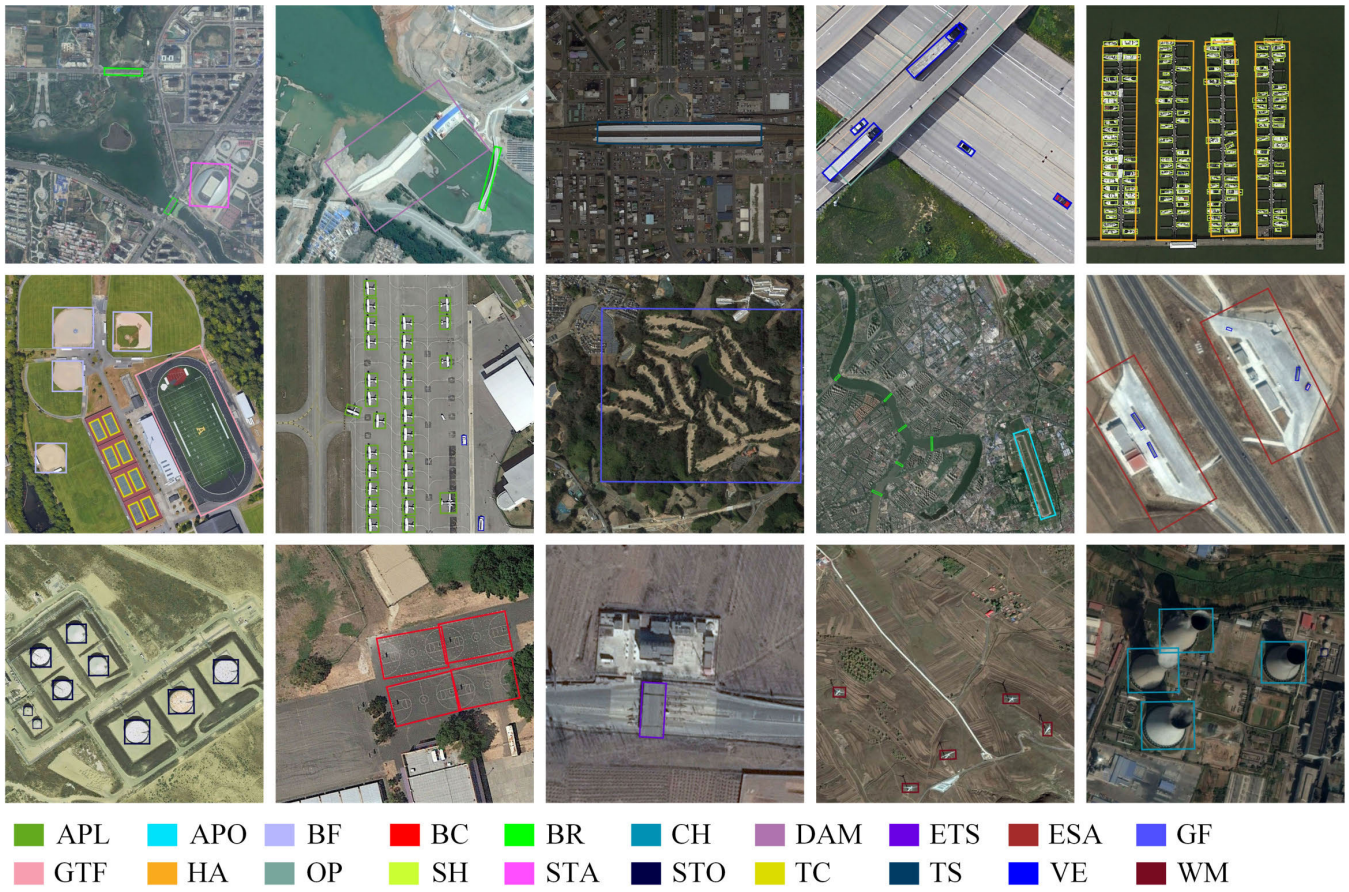


Fig. 6. Some detection results of TCD on the DIOR-R dataset.

TABLE V
COMPARISON WITH METHODS ADOPTING DECOUPLED HEADS

Methods	Image Size	AP50	AP75
TSD [47]	1024×1024	66.83	38.32
Double Head R-CNN [48]	1024×1024	67.02	38.76
YOLOX-s [49]	640×640	63.75	32.10
YOLOX-x [49]	640×640	68.23	39.58
TCD	1024×1024	**69.86**	**41.30**

TABLE VI
ANALYSIS OF DIFFERENT α AND β FOR TCA

β	α	AP50	β	α	AP50	β	α	AP50
1	0.2	69.35	2	0.2	69.79	3	0.2	68.91
	0.3	69.39		0.3	69.73		0.3	68.66
	0.4	68.86		0.4	**69.86**		0.4	68.39
	0.5	69.67		0.5	**69.86**		0.5	68.42
	0.6	69.11		0.6	69.84		0.6	67.94

TABLE VII
ANALYSIS OF DIFFERENT k FOR TCA

k	4	6	8	10	12	15
AP50	68.75	69.78	**69.86**	69.75	69.56	69.63

TABLE VIII
COMPUTATIONAL EFFICIENCY ANALYSIS OF TCD. N1 AND N2 REPRESENT THE NUMBER OF FEATURE AGGREGATION LAYERS AND CLASSIFICATION/LOCALIZATION LAYERS, RESPECTIVELY. FLOPS AND RUNNING TIME ARE MEASURED ON THE INPUT IMAGE SIZE OF 1024 × 1024 AND A GTX2080TI

Method	N1	N2	Head params	FLOPs	Running time	AP50
TCD	1	1	0.5 M	109.23 G	0.0500 s	57.01
		2	1.7 M	134.97 G	0.0511 s	61.55
		3	2.9 M	160.72 G	0.0516 s	63.23
		4	4.0 M	186.46 G	0.0528 s	66.56
	2	1	0.8 M	116.77 G	0.0505 s	59.51
		2	2.0 M	142.52 G	0.0513 s	62.95
		3	3.2 M	168.26 G	0.0523 s	66.71
		4	4.4 M	194.01 G	0.0532 s	67.85
	4	1	1.5 M	131.86 G	0.0511 s	64.21
		2	2.7 M	157.60 G	0.0519 s	68.75
		3	3.9 M	183.35 G	0.0531 s	**69.86**
		4	5.1 M	209.09 G	0.0542 s	69.85
RetinaNet-O			4.8 M	202.25 G	0.0520	62.89

and 7.62 AP75 over the baseline (IoU-based). To be more specific, TCA outperforms the current assignment methods such as ATSS (by 2.03 AP50), DAL (by 1.06 AP50), and PAA (by 0.85 AP50). These validate that the assignment scheme of task-collaborated can improve the detection performance

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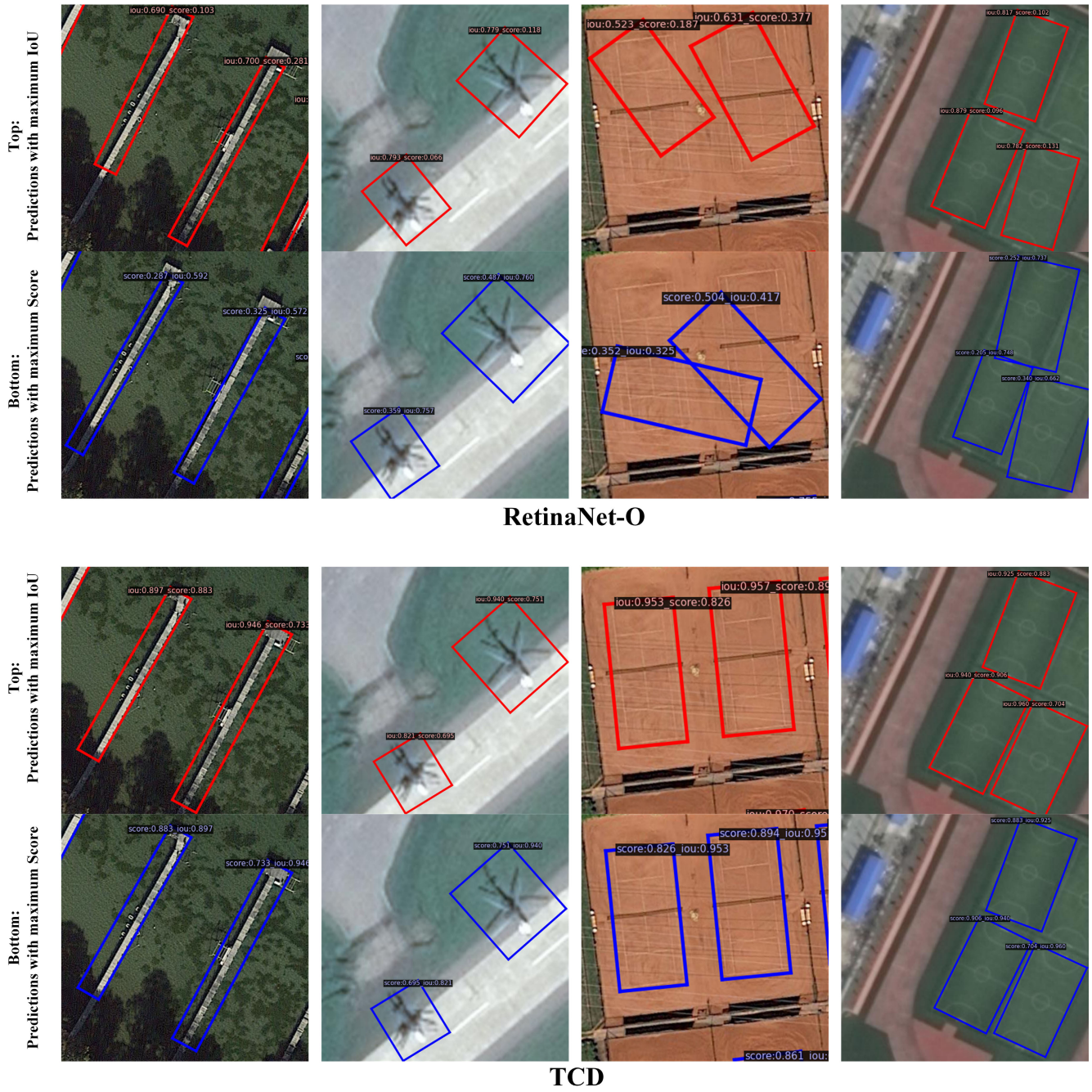


Fig. 7. Some detection results in RetinaNet-O and TCD. The red and blue represent the prediction boxes with the highest localization confidence and classification confidence, respectively. TCD is significantly better than RetinaNet-O in both prediction accuracy and consistency.

effectively. Notably, when TCA and TCH are combined, TCD brings a higher improvement (5.86 AP50 above baseline) than ATSS + TCH (3.98 AP50 above baseline), DAL + TCH (4.51 AP50 above baseline), and PAA + TCH (4.92 AP50 above baseline). It suggests that TCA and TCH can collaboratively bring higher performance. Finally, TCL can also bring more than one AP50 performance improvement by introducing a soft IoU label.

2) Comparison With Methods Adopting Decoupled Heads: Due to the different features required for classification and localization, there are some detectors that employ different head structures for two different tasks rather than two parallel branches and have been successful, i.e., classification uses fully connected layers and localization uses convolutional layers. Unlike them, TCD pays more attention to the interaction of the two branches of the head and aims to improve the consistency of the predictions of the two tasks (high classification score with high localization confidence). Table V shows the comparison between TCD and three methods with decoupled head. They all add angle prediction based on mmdetection,[5] where TSD and Double Head R-CNN use resnet50 as the backbone, and YOLOX uses S and X versions.

[5]https://github.com/open-mmlab/mmdetection

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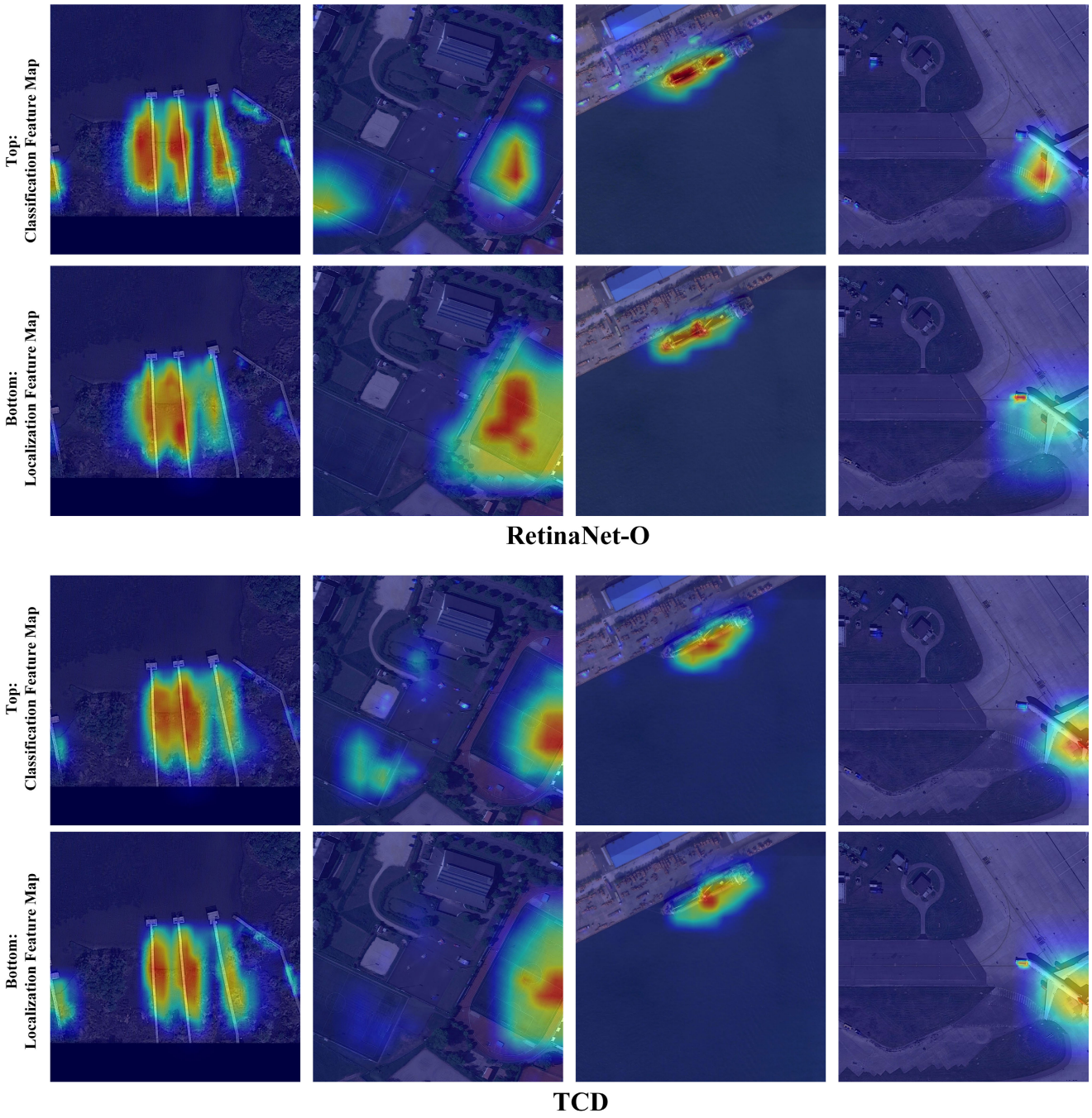


Fig. 8. Visualization of classification and localization feature maps in RetinaNet-O and TCD. Red indicates that features are more activated. The activation positions of the classification and localization features of TCD tend to be more consistent.

It can be seen that TCD has a stable performance improvement (1.63–3.03 AP50), which also shows that the structure of TCD is more advantageous to a certain extent.

3) Hyperparameters: to find suitable hyperparameters and explore the impact of input IoU and prediction on sample assignment, we first conduct the experiments using different α and β for TCA, which can control the proportion of two tasks in the assignment metric. As shown in (1) and (2), the larger the β, the greater the influence of input IoU on the assignment metric, and on the contrary, the smaller the influence of the two tasks' prediction. The results in Table VI indicate that too large or too small β is inappropriate, TCD has a better performance when $\beta = 2$ and we adopt $\alpha = 0.5$ and $\beta = 2$ for the TCA. Then we conduct experiments for the top k with different values in [4–15], which is used to determine the number of positive samples. The results in Table VII indicate that TCD has a similar performance when k is in the above interval ($k = 8$ is the best), which shows that TCD is not sensitive to k.

4) Computational Efficiency Analysis of TCD: As shown in Table VIII, We compare the head parameters, floating point operations, and algorithm running time under different

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TABLE IX
CONSISTENCY ANALYSIS OF TCD, TOOD-R INDICATES THAT THE ANGLE PREDICTION IS ADDED ON THE BASIS OF TOOD [51], AND OTHER SETTINGS ARE THE SAME AS ORIGINAL PAPER. DAS AND MSL REPRESENT THE DYNAMIC ANCHOR SELECTION AND MATCHING-SENSITIVE LOSS IN DAL [45]

Method	AP50	Precision	Recall	F1	PCC (top 8)	IoU (top 8)	CC (top 1)
RetinaNet-O	62.89	58.39	64.28	61.19	0.392	0.672	0.292
ATSS [5] + TCH	66.87	64.66	67.28	65.94	0.437	0.725	0.323
DAS [42] +TCH	67.40	63.25	68.87	65.94	0.474	0.757	0.358
DAS [42] +TCH +MSL [42]	68.70	64.23	69.01	66.53	0.526	0.799	0.375
TCA+TCH	68.75	65.56	68.24	66.87	0.513	0.792	0.386
TOOD-R [51]	69.63	66.69	68.75	67.70	0.539	0.807	0.413
TOOD-R [51] + TCA	69.70	66.83	**69.21**	68.00	0.551	**0.819**	0.415
TCA+TCH+TCL	**69.86**	**67.01**	69.16	**68.07**	**0.563**	0.810	**0.421**

N1 and N2. N1 and N2 represent the number of feature aggregation layers and classification/localization layers, respectively. When both N1 and N2 are equal to 2, TCD has a similar performance to RetinaNet-O, with fewer parameters and running time. When N1 is equal to 4 and N2 is equal to 3, compared to RetinaNet-O, TCD has higher performance (+6.97 AP50), fewer parameters (−0.9 M), fewer floating point operations (−18.9 G), and approximate running time (+0.0011 s).

5) Consistency Analysis of TCD: As shown in Table IX, we conducted a quantitative analysis for the impact of TCD on the two tasks' output consistency. First, we compare the precision, recall, and F1 score of TCD with other state-of-the-art methods based on a classification confidence threshold of 0.3. The results in Table IX show that TCD has the highest accuracy and F1 score. Then, we selected the top eight predictions with the highest localization confidence for each object before NMS and calculated the PCC between localization confidence and corresponding classification confidence. TCD achieves 0.171 improvements over RetinaNet-O and has 0.037–0.126 improvements compared to other advanced algorithms. The correlation graph between the localization and classification score of RetinaNet-O and TCD is shown in Fig. 2. Moreover, mean IoU of the top eight predictions is calculated, TCD still achieves a great (0.138) improvement over RetinaNet-O. Finally, the consistency coefficients (CCs) [defined as (15), arg max represent the index of the max value, N denotes the number of objects] between the best localization and classification score in the top eight predictions are calculated to reflect the consistency of the optimal prediction of the two tasks. TCD achieves 0.129 higher than RetinaNet-O, and it is also better than other advanced methods, as shown in Table IX. At the same time, it is not difficult to find that there is still much promotion space for optimization in the consistency of the two tasks' predictions. Several detection examples with the most outstanding classification or localization confidence are illustrated in Fig. 7. It can also be seen from the feature map in Fig. 8 that the feature activations of classification and localization in TCD tend to be more consistent

$$\mathrm{CC} = \frac{\sum_{i=1}^{N} T_i(\arg\max(l), \arg\max(c))}{N}$$
$$T(a,b) = \begin{cases} 1, & a = b \\ 0, & a \neq b. \end{cases} \tag{15}$$

V. CONCLUSION

This article analyzes the prediction inconsistency problem of classification and localization from two aspects and proposes TCD to alleviate this problem. The proposed TCA and TCH push the network to select higher quality anchors and make the two tasks more interactive. Moreover, a CC is proposed to measure the consistency relationship between the predictions of two tasks. With the improvements of TCA and TCH, TCD achieves the state-of-the-art performance on DOTA, DIOR-R, and HRSC2016 and dramatically improves the prediction consistency. In future work, we will explore more advanced label assignment methods, try to remove the complex postprocessing process, and build a more elegant and concise end-to-end OOD pipeline.

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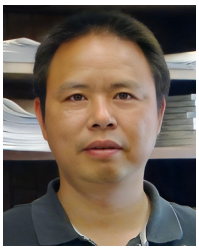
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